

Data-driven (arsenic) contamination detection

Description:

Utilize the provided starter code for creating arsenic contaminations, which are reacting with chlorine, to build a data set for arsenic contamination event detection – i.e., a large number of scenarios with random contamination events. You may use a simple chlorine injection pattern, such as a constant injection. Perform a literature search (starting with the provided material) on event/anomaly detection methods in general and for water contamination specifically. Pick one method and implement it. The method must only use flow and chlorine concentration sensors. Design an evaluation method and use it to evaluate the detection method. Investigate what sizes and locations of contamination can be detected well.

Possible extensions to group projects:

- Use other benchmark networks (e.g., Hanoi) for building the dataset
- Implement and evaluate different event/anomaly/contamination detection methods
- Include other events such as sensor faults, leakages, etc., in the data set
- Investigate the performance of detection methods on the wastewater contamination events of the [“1st AI for Drinking Water Chlorination Challenge @ IJCAI-25”](#)
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